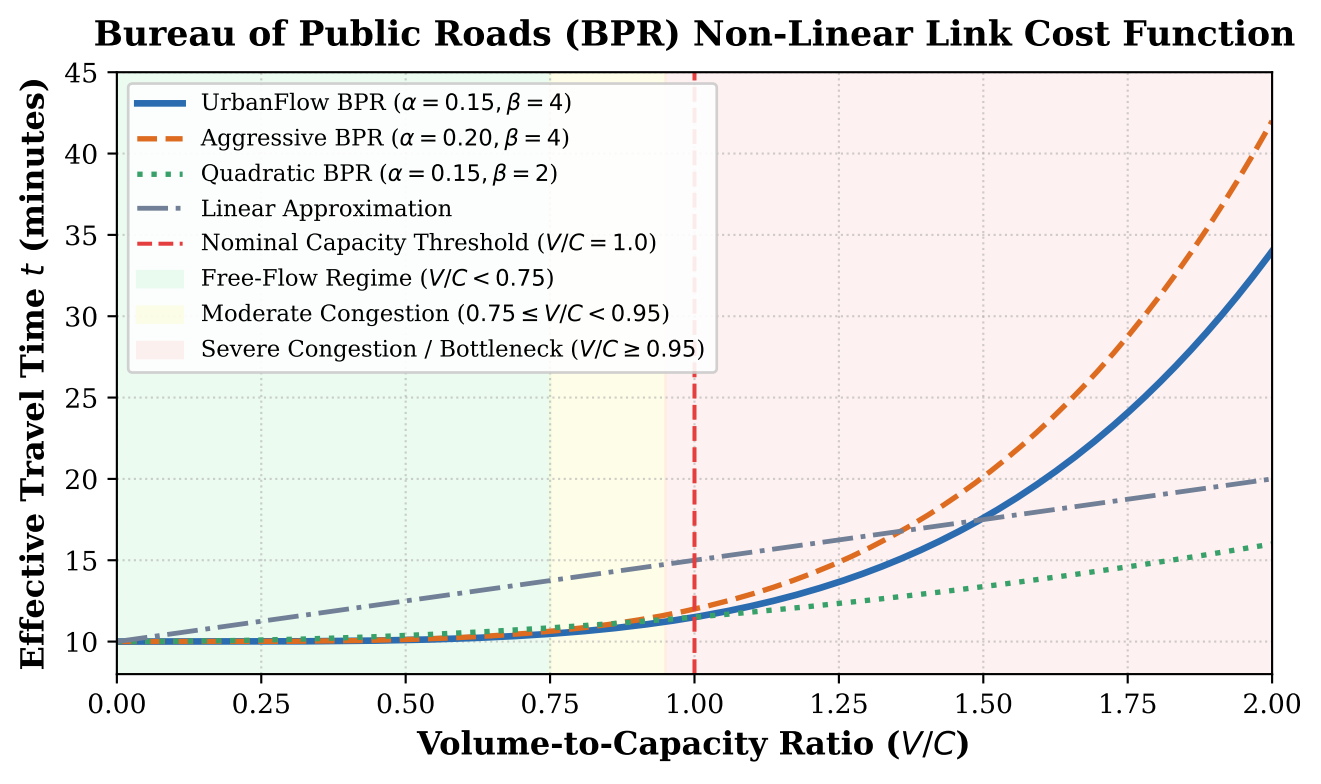
Bureau of Public Roads (BPR) Non-Linear Link Cost Function
Effective Travel Time t (minutes)
Volume-to-Capacity Ratio (V/C)
UrbanFlow BPR ($\alpha = 0.15, \beta = 4$)
Aggressive BPR ($\alpha = 0.20, \beta = 4$)
Quadratic BPR ($\alpha = 0.15, \beta = 2$)
Linear Approximation
Nominal Capacity Threshold ($V/C = 1.0$)
Free-Flow Regime ($V/C < 0.75$)
Moderate Congestion ($0.75 \leq V/C < 0.95$)
Severe Congestion / Bottleneck ($V/C \geq 0.95$)
45
40
35
30
25
20
15
10
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00